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Smudge detection system and its smudge detection device

Abstract

A smudge detection device includes a container, a power supply battery, a smudge detection module, a photovoltaic device and a mainboard module. The container is formed with a recess at the top thereof, and a light transmissive cover covering the recess. The smudge detection module is located in the recess to detect the shading degree of the light transmissive cover that is covered. The photovoltaic device is disposed on the container to generate and transmit electric energy into the power supply battery. The mainboard module is electrically connected to the smudge detection module, the power supply battery and the photovoltaic device, and connected to a host server through network architecture, so as to respectively transmit power supply value of the electric energy and the shading degree of the light transmissive cover to the host server.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0338771	12/2016	Gostein et al.	N/A	N/A
2018/0331654	12/2017	Gostein	N/A	G01N 21/9501

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
213072559	12/2020	CN	N/A
1285699	12/2021	ES	N/A
M609396	12/2020	TW	N/A
M623556	12/2021	TW	N/A
202219788	12/2021	TW	N/A
202222030	12/2021	TW	N/A

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Background/Summary**BACKGROUND****Field of Disclosure**

(1) The present disclosure relates to a detection system. More particularly, the present disclosure relates to a smudge detection system and its smudge detection device.

Description of Related Art

(2) Generally, surfaces of many objects (e.g., telescope lens, building glass or solar panels, etc.) need to be cleaned regularly by personnel to maintain their original appearance or service performance. For example, when current solar power plants are built, multiple solar modules are usually arranged side by side in order, and then the electric energy generated by the solar modules is input into the mains system and resold to electricity selling units.

(3) However, since solar energy sites are located outdoors for a long time, dust accumulated on the solar modules, sludge mixed by dust and rain, or snow or frost in cold weather will affect the photoelectric conversion efficiency of the solar modules. Currently, a solution that is taken so far is to regular inspection manually for cleaning the smudge on the light incident surface of the solar modules. In this way, since the progress of pollution is very difficult to predict, it is necessary to increase inspector shifts, thereby increasing labor costs and cleaning costs.

(4) Thus, the above-mentioned technology obviously still has inconveniences and defects, which are issues that the industry needs to solve urgently.

SUMMARY

(5) One aspect of the present disclosure is to provide a smudge detection system and its smudge

detection device for solving the difficulties mentioned above in the prior art.

(6) In one embodiment of the present disclosure, a smudge detection device that is provided includes a container, a power supply battery, a smudge detection module, a photovoltaic device and a mainboard module. The container includes a main body disposed on at least one power generation module, a recess formed on a top of the main body, and a light transmissive cover covering the recess. The smudge detection module is located within the recess for detecting a shielding degree of the light transmissive cover being covered by foreign matters. The photovoltaic device is disposed on the main body for generating and transmitting electric energy into the power supply battery. The mainboard module is received within the recess, and includes a communication unit connected to a host server through a network architecture, and a control unit electrically connected to the smudge detection module, the power supply battery, the communication unit and the photovoltaic device, for respectively transmitting power supply value of the electric energy and the shielding degree of the light transmissive cover to the host server.

(7) According to one or more embodiments of the present disclosure, in the smudge detection device, the light transmissive cover is a single light-transmissive plate, and the single light-transmissive plate is divided into a first area and a second area. The photovoltaic device is directly integrated into the first area of the single light-transmissive plate, and the smudge detection module is faced towards the second area of the single light-transmissive plate from inside the recess for detecting the shielding degree of the second area being covered.

(8) According to one or more embodiments of the present disclosure, in the smudge detection device, the smudge detection module includes at least one outputting unit and a receiving unit. The outputting unit is disposed within the recess, and electrically connected to the control unit for sending energy waves towards the light transmissive cover. The receiving unit is disposed within the recess, and electrically connected to the control unit for receiving reflected signals returned from the light transmissive cover for the control unit to transmit the reflected signals to the host server.

(9) According to one or more embodiments of the present disclosure, in the smudge detection device, the outputting unit includes a plurality of outputting units, and the outputting units are spaced arranged to surround the receiving unit.

(10) According to one or more embodiments of the present disclosure, in the smudge detection device, the output unit is a light emitter, and the receiving unit is a light sensor, or the output unit is an ultrasonic transmitter, and the receiving unit is an ultrasonic receiver.

(11) According to one or more embodiments of the present disclosure, in the smudge detection device, the smudge detection module includes a camera unit disposed within the recess, and electrically connected to the control unit for capturing images of the light transmissive cover for the control unit to transmit the images to the host server.

(12) According to one or more embodiments of the present disclosure, in the smudge detection device, the mainboard module further includes a global positioning system (GPS) unit electrically connected to the control unit for obtaining location coordinates where the container is situated at, so that the control unit transmits the location coordinates of the container to the host server.

(13) According to one or more embodiments of the present disclosure, in the smudge detection device, the GPS unit further obtains an absolute time where the container is situated at, so that the control unit is able to activate the smudge detection module based on the absolute time of the container.

(14) In one embodiment of the present disclosure, a smudge detection system is suitable for determining whether a surface of an observed object is smudgy, and the smudge detection system includes a host server and a smudge detection device. The smudge detection device includes a power supply battery, a container, a smudge detection module, a photovoltaic device and a mainboard module. The container includes a main body disposed on the observed object, a recess formed on a top of the main body, and a light transmissive cover covering the recess. The smudge

detection module is located within the recess for detecting a shielding degree of the light transmissive cover being covered by foreign matters. The photovoltaic device is disposed on the main body for generating and transmitting electric energy into the power supply battery. The mainboard module is received within the recess, electrically connected to the smudge detection module, the power supply battery and the photovoltaic device, and connected to a host server through a network architecture, for respectively transmitting power supply value of the electric energy and the shielding degree of the light transmissive cover to the host server. The host server determines whether the shielding degree of the light transmissive cover exceeds a preset threshold, and whether the power supply value of the photovoltaic device is within a predetermined range. When the host server determines that the shielding degree exceeds the preset threshold, and the power supply value of the photovoltaic device is not within the predetermined range, the host server issues a smudge-detected alert outwardly.

(15) According to one or more embodiments of the present disclosure, in the smudge detection system, the light transmissive cover is a single light-transmissive plate, and the single light-transmissive plate is divided into a first area and a second area. The photovoltaic device is directly integrated into the first area of the single light-transmissive plate, and the smudge detection module faces towards the second area of the single light-transmissive plate from inside the recess for detecting the shielding degree of the second area being covered.

(16) According to one or more embodiments of the present disclosure, in the smudge detection system, the smudge detection module includes at least one outputting unit and a receiving unit. The outputting unit is disposed within the recess, and electrically connected to the mainboard module for sending energy waves towards the light transmissive cover. The receiving unit is disposed within the recess, and electrically connected to the mainboard module for receiving reflected signals returned from the light transmissive cover. The mainboard module transmits the reflected signals to the host server, so that the host server is able to compare a difference between the reflected signal and the preset threshold.

(17) According to one or more embodiments of the present disclosure, in the smudge detection system, the outputting unit includes a plurality of outputting units, and the outputting units are spaced arranged to surround the receiving unit.

(18) According to one or more embodiments of the present disclosure, in the smudge detection system, the output unit is a light emitter, and the receiving unit is a light sensor, or the output unit is an ultrasonic transmitter, and the receiving unit is an ultrasonic receiver.

(19) According to one or more embodiments of the present disclosure, in the smudge detection system, the smudge detection module includes a camera unit disposed within the recess, and electrically connected to the mainboard module for capturing images of the light transmissive cover for the mainboard module to transmit the images to the host server.

(20) According to one or more embodiments of the present disclosure, in the smudge detection system, when the host server determines whether the power supply value of the photovoltaic device is within the predetermined range, the host server further proceeds to collect an ideal power supply value of the photovoltaic device, to compare the power supply value of the photovoltaic device with the ideal power supply value to obtain a difference value, to determine whether the difference value is in the predetermined range, and to issue the smudge-detected alert outwardly when the difference value is not within the predetermined range.

(21) According to one or more embodiments of the present disclosure, in the smudge detection system, the mainboard module further includes a global positioning system (GPS) unit electrically connected to the mainboard for obtaining location coordinates where the container is situated at, so that the mainboard transmits the location coordinates of the container to the host server.

(22) When the host server determines whether the power supply value of the photovoltaic device is within the predetermined range, the host server further proceeds to obtain weather data and real-time sunshine data based on the location coordinates of the container, to estimate an expected

power supply value based on the weather data and the real-time sunshine data, to compare the power supply value and the expected power supply value of the photovoltaic device to obtain a difference value, and to issue the smudge-detected alert outwardly when the difference value is not within the predetermined range.

(23) Thus, through the construction of the embodiments above, the disclosure is able to accurately decide whether an observed object is smudgy, so as to further timely send labor to eliminate abnormalities, thereby stabilizing inspector shift time and saving labor and cleaning costs.

(24) The above description is merely used for illustrating the problems to be resolved, the technical methods for resolving the problems and their efficacies, etc. The specific details of the present disclosure will be explained in the embodiments below and related drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

(2) FIG. 1 is a schematic view of a smudge detection system according to one embodiment of the present disclosure.

(3) FIG. 2 is a schematic view of the smudge detection device in FIG. 1.

(4) FIG. 3 is a block diagram of the smudge detection device in FIG. 1.

(5) FIG. 4 is a schematic view of a smudge detection device according to one embodiment of the present disclosure.

(6) FIG. 5 is a schematic view of a smudge detection system implemented on a solar energy site according to one embodiment of the present disclosure.

(7) FIG. 6 is a flowchart of a smudge detection method according to one embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

(8) Reference will now be made in detail to the present embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts. According to the embodiments, it will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure.

(9) Reference is now made to FIG. 1 to FIG. 3, in which FIG. 1 is a schematic view of a smudge detection system **10** according to one embodiment of the present disclosure, FIG. 2 is a schematic view of the smudge detection device **300** in FIG. 1, and FIG. 3 is a block diagram of the smudge detection device **300** in FIG. 1. As shown in FIG. 1 to FIG. 3, in this embodiment, the smudge detection system **10** is suitable for determining whether a surface of an observed object **D** is contaminated (e.g., smudgy), and the smudge detection system **10** includes a host server **100** and at least one smudge detection device **300**. The host server **100** is connected to the smudge detection device **300** through a network architecture **N**. The smudge detection device **300** is physically disposed on the aforementioned observed object **D** to detect whether the surface of the aforementioned observed object **D** is contaminated (e.g., smudgy).

(10) It is noted, the disclosure is not limited to the type of the aforementioned observed object **D**. For example, the aforementioned observed object **D** is telescope lens, building glass or solar panels, etc.

(11) The smudge detection device **300** includes a container **310**, a power supply battery **600**, a

smudge detection module **400**, a photovoltaic device **500** and a mainboard module **700**. The container **310** includes a main body **320**, a recess **330** and a light transmissive cover **350**. The main body **320** is arranged aside the aforementioned observed object D. The recess **330** is formed on a top **321** of the main body **320**. The light transmissive cover **350** covers the recess **330**, and further totally covers the recess **330** of the container **310**. The smudge detection module **400** is located within the recess **330** and facing towards the light transmissive cover **350** for performing smudge detection which is to detect a shielding degree of the light transmissive cover being covered by foreign matters within the recess **330**. The photovoltaic device **500** is arranged aside the aforementioned observed object D. In the embodiment, the photovoltaic device **500** is single and disposed on the main body **320**. The mainboard module **700** is electrically connected to the smudge detection module **400**, the power supply battery **600** and the photovoltaic device **500**, and connected to the host server **100** through the network architecture N for respectively transmitting power supply value of the electric energy and the shielding degree of the light transmissive cover **350** to the host server **100**. In this embodiment, the power supply battery **600** is located within the container **310**, and the photovoltaic device **500** is located within the recess **330** to generate and transmit electrical energy to the power supply battery **600**. The power supply battery **600** is able to supply working power to various components of the smudge detection device **300**, however, the disclosure is not limited thereto.

(12) Thus, the host server **100** determines whether the shielding degree of the light transmissive cover **350** exceeds a preset threshold, and whether the power supply value of the photovoltaic device **500** is within a predetermined range. When the host server **100** determines that the shielding degree exceeds the preset threshold, and the power supply value of the photovoltaic device **500** is not within the predetermined range, the host server **100** issues a smudge-detected alert (e.g., text message, email or other audio and video signal) outwardly to a terminal T (e.g., mobile phone, computer or other cloud server).

(13) Thus, through the construction of the embodiments above, the disclosure is able to accurately decide whether an observed object D is smudgy, so as to further timely send labor to eliminate abnormalities, thereby stabilizing inspector shift time and saving labor and cleaning costs.

(14) More specifically, in this embodiment, the power supply battery **600**, the mainboard module **700** and the smudge detection module **400** are all fixedly disposed at a bottom **331** of the recess **330**, and the power supply battery **600**, the smudge detection module **400** and the mainboard module **700** are spaced apart from each other. The mainboard module **700** is located between the power supply battery **600** and the smudge detection module **400**, and electrically connected to the photovoltaic device **500** and the smudge detection module **400** through wires C respectively.

(15) Furthermore, the mainboard module **700** includes a circuit board **710**, a communication unit **720**, a storage unit **730** and a control unit **750**. The circuit board **710** is secured to the bottom **331** of the recess **330** through some studs B, and the communication unit **720**, the storage unit **730** and the control unit **750** are respectively soldered to the circuit board **710**. The communication unit **720** is connected to the host server **100** through the network architecture N. The storage unit **730** is stored with any numerical information or temporary data transmitted from the smudge detection module **400** and the photovoltaic device **500**, as well as control programs for performing specific tasks and work. The control program is not limited to the form of software, firmware and/or programs. The control unit **750** is electrically connected to the smudge detection module **400**, the power supply battery **600**, the communication unit **720**, the storage unit **730** and the photovoltaic device **500** for respectively transmitting power supply value of the electric energy and the shielding degree of the light transmissive cover **350** to the host server **100**.

(16) The communication unit **720** is electrically connected to the host server **100** through a wireless manner, for example, through an antenna unit **760** to be wirelessly connected to the host server **100**. The communication unit **720** can support several communication protocols such as Global System for Mobile communication (GSM), Personal Handy-phone System (PHS), Code Division Multiple

Access (CDMA), Broadband Code Wideband Code Division Multiple Access (WCDMA), Long Term Evolution (LTE), Worldwide interoperability for Microwave Access (WiMAX), Wireless Fidelity (Wireless Fidelity, Wi-Fi) and Bluetooth etc., but the present disclosure is not limited thereto.

(17) In addition, the mainboard module **700** further includes a global positioning system (GPS) unit **740**. The GPS unit **740** is soldered on the circuit board **710** and electrically connected to the control unit **750** for obtaining absolute time (i.e., location time of the geographical location) and location coordinates where the smudge detection device **300** (i.e., the container **310** or the aforementioned observed object D) is situated at from the satellite signal through the antenna unit **760**, and then the control unit **750** transmits the location coordinates to the host server **100**. Since the smudge detection module **400** is not needed to perform smudge detection around the clock, when a setup time is up, by comparing the absolute time and the setup time, the control unit **750** immediately executes the activation and deactivation of the smudge detection module **400** through the control program stored in the storage unit **730** so as to save energy and decrease the loss of product life.

(18) For example, the communication unit **720** is not limited to various communication chips, mobile communication chips, Bluetooth chips, Long Term Evolution (LTE) chips or WiFi chips, etc.; the control unit **750** is a central processing unit (CPU), single chip or microcontroller; the storage unit **730** is a removable random access memory, flash memory or hard disk, etc.; the GPS unit **740** is not limited to various types of GPS chips, and the GPS unit **740** supports, for example, global navigation satellite systems (Global Navigation Satellite System, GNSS) and other protocols, the network architecture N is not limited to the Internet, WIFI networks, Intranet networks or Extranet networks belonging to the same enterprise or organization, etc., however, the disclosure is not limited to the above-mentioned aspects.

(19) Also, the smudge detection module **400** includes a wire board **410**, a receiving unit **420** and a plurality of outputting units **430**. The wire board **410** is secured to the bottom **331** of the recess **330** through some studs B, and the receiving unit **420** and the outputting units **430** are respectively soldered to the wire board **410**, and the outputting units **430** are spaced arranged to surround the receiving unit **420**. The outputting units **430** are electrically connected to the control unit **750** for sending energy waves towards the light transmissive cover **350**, and the energy waves might be partially reflected back by the smudge on the light transmissive cover **350**. The receiving unit **420** is electrically connected to the control unit **750** for receiving reflected signals returned from the light transmissive cover **350** for the control unit to transmit the reflected signals to the host server **100**.

(20) In this embodiment, for example, each of the outputting units **430** is a light emitter (e.g., light emitting diode or infrared diode), and the receiving unit **420** is a light sensor (e.g., photodiode). In this way, when the light emitter emits light L1 towards the light transmissive cover **350**, one part of the light L1 penetrates through the light transmissive cover **350**, and another part of the light L1 is reflected by the light transmissive cover **350** and returns to the smudge detection module **400**. Therefore, when the light sensor receives the reflected light L2 returned from the light transmissive cover **350**, the light sensor converts the reflected light L2 into a photoelectric signal. However, the light of the light emitter is not limited to be the visible light or invisible light.

(21) Therefore, after the photoelectric signal is transmitted to the host server **100**, the host server **100** compares the difference between the photoelectric signal and the preset threshold to determine whether the outer surface of the light transmissive cover **350** and the aforementioned observed object D is smudgy so as to decide whether a smudge-detected alert is issued outwardly.

(22) It is noted, when one part of the light transmissive cover **350** is covered by foreign matter (e.g., dust, pollen, poultry excrement or a combination thereof), compared to the cleaner part of the light transmissive cover **350**, the light sensor will receive stronger photoelectric signals. In this way, according to the difference in photoelectric signals, the smudge detection module **400** can decide the shielding degree of the light transmissive cover **350** being covered by foreign matters.

(23) However, the disclosure is not limited to this. In other embodiments, the outputting units **430** of the smudge detection module **400** also can be replaced by an ultrasonic transmitter, and the receiving unit **420** also can be replaced by an ultrasonic receiver.

(24) Furthermore, a supporting flange **340** is protruded from an opening of the recess **330**, and the supporting flange **340** surrounds a receiving groove **341** for loading the light transmissive cover **350**. The light transmissive cover **350** is placed flat within the receiving groove **341** so that the light transmissive cover **350** is totally embedded within the receiving groove **341**, and the outer surface of the light transmissive cover **350** is flush with the top **321** of the container **310**. In addition, a portion of the supporting flange **340** is formed with a receiving groove **342** for receiving the photovoltaic device **500**.

(25) Furthermore, the light transmissive cover **350** is a single light-transmissive plate **351** (e.g., glass plate or acrylic plate). The single light-transmissive plate **351** is divided into a first area **351A** and a second area **351B** along a horizontal direction, and the first area **351A** and the second area **351B** are adjacent to each other, and there is no physical boundary between the first area **351A** and the second area **351B**. The photovoltaic device **500** is only located in the first area **351A** of the single light-transmissive plate **351** and directly integrated into one side of the first area **351A** facing towards the mainboard module **700**. Furthermore, the photovoltaic device **500** is directly integrated into one surface (hereinafter referred to as the inner surface) of the first area **351A** of the single light-transmissive plate **351** facing towards the mainboard module **700**, and merely arranged on the first area **351A**. In other words, the area of the single light-transmissive plate **351** covered by the photovoltaic device **500** is the first area **351A** of the single light-transmissive plate **351**, and the other thereof not covered by the photovoltaic device **500** is the second area **351B**, however the disclosure is not limited thereto. The smudge detection module **400** is faced towards the second area **351B** of the single light-transmissive plate **351** for detecting the shielding degree of the second area **351B** being covered by foreign matters.

(26) More specifically, the photovoltaic device **500** includes a photovoltaic panel **510** and a junction box **520**. The photovoltaic panel **510** is placed flat in the receiving groove **342** and directly attached to the inner surface of the single light-transmissive plate **351**. The junction box **520** is located on one side of the supporting flange **340** facing away from the receiving groove **342**, and electrically connected to the photovoltaic panel **510**, the power supply battery **600** and the mainboard module **700** for transmitting the electric energy generated by the photovoltaic panel **510** to the power supply battery **600**, and transmitting the power supply value of the electric energy to the mainboard module **700**.

(27) FIG. 4 is a schematic view of a smudge detection device **301** according to one embodiment of the present disclosure. As shown in FIG. 4, the smudge detection device **301** of the present embodiment is substantially the same to the smudge detection device **300** in FIG. 1, and the difference therebetween is that, the receiving unit **420** and the outputting units **430** in the smudge detection module **401** of this embodiment can also be replaced with a camera unit **440**. The camera unit **440** is disposed within the recess **330**, and soldered to the wire board **410**. The camera unit **440** is electrically connected to the control unit **750** for capturing images of the light transmissive cover **350** for the control unit **750** to transmit the images to the host server **100**.

(28) It is noted, when one part of the light transmissive cover **350** is covered by foreign matter (e.g., dust, pollen, poultry excrement or a combination thereof), compared to the cleaner part of the light transmissive cover **350**, the image captured by the camera unit **440** will with more shadows. In this way, depending on the amount of shadows in the image, the smudge detection module **400** can decide the shielding degree of the light transmissive cover **350** being covered by foreign matters.

(29) FIG. 5 is a schematic view of a smudge detection system **11** implemented on a solar energy site according to one embodiment of the present disclosure. As shown in FIG. 5, the smudge detection device **302** of the present embodiment is substantially the same to the smudge detection

device **300** in FIG. **1**, and the difference therebetween is that, the smudge detection system **11** is implemented on a solar energy site, and the solar energy site includes a solar array **200**. The solar array **200** is connected to the host server **100** through the network architecture N. The solar array **200** includes a frame **210**, an inverter **220** and a plurality of power generation modules **230**. Each of the power generation modules **230** is a solar module and arranged on the frame **210**. The power generation modules **230** are arranged in an array. The inverter **220** is electrically connected to the power generation modules **230** respectively, and is connected to the host server **100** through the network architecture N, so as to receive and output the power supply values of the power generation modules **230** to the host server **100**. In this way, when the host server **100** issues a smudge-detected alert outwardly, the controller can be programmed to decide that the light incident surface of the adjacent power generation module **230** is also polluted (e.g., smudgy) to the same extent, and thus, subsequent cleaning procedures can be then performed.

(30) However, the disclosure is not limited to this detection mechanism. In other embodiments, when the host server **100** determines that the surface of the light transmissive cover **350** is smudgy, the host server **100** continues to compare the power generation of the solar array **200**. When the power supply value of the power generation module **230** is approximately the same as the power supply value of the photovoltaic device **500**, the host server **100** will issue a smudge-detected alert outwardly. Therefore, it can be inferred that the light incident surface of the adjacent power generation module **230** is also polluted (e.g., smudgy) to the same extent, and subsequent cleaning procedures can be then performed.

(31) FIG. **6** is a flowchart of a smudge detection method according to one embodiment of the present disclosure. As shown in FIG. **6**, the smudge detection method includes step **801** to step **805** described below. In the step **801**, energy waves are sent towards a light transmissive cover and reflected signals returned from the light transmissive cover are received so as to obtain a shielding degree of the light transmissive cover covered by foreign matters. In the step **802**, a determination is made as to whether the shielding degree of the light transmissive cover exceeds a preset threshold, if yes, go to step **803**, otherwise back to step **801**. In the step **803**, a power supply value of the electric energy generated by the photovoltaic device is obtained. In the step **804**, a determination is made as to whether the power supply value of the electric energy generated by the photovoltaic device is within a predetermined range, if yes, back to step **803**, otherwise go to step **805**. In the step **805**, a smudge-detected alert is issued outwardly.

(32) However, the disclosure is not limited thereto. In other embodiments, the order of steps **801** to **802** and steps **803** to **804** may also be performed alternately.

(33) More specifically, in a detailed option of this embodiment, in step **804**, the host server **100** processes according to detailed steps as followed.

(34) First, the host server **100** collects an ideal power supply value of the photovoltaic device **500** (e.g., the recent best performance after installation); then, the host server **100** compares the power supply value of the photovoltaic device **500** with the ideal power supply of the photovoltaic device **500**, and obtains a difference value between the power supply value and the ideal power supply; next, a determination is made as to whether the difference value is in the predetermined range above, if so, back to step **803**, otherwise, continue to step **805**.

(35) More specifically, in a detailed option of this embodiment, in step **804**, as shown in FIG. **1** to FIG. **3**, the host server **100** proceeds according to the following detailed steps. First, the host server **100** obtains the position coordinates of the solar energy site. For example, the host server **100** obtains the position coordinates and absolute time through the GPS unit **740**; then, the host server **100** obtains weather data and real-time sunshine data of the location of the smudge detection device **300** (i.e., the container **310** or the observed object D) according to the position coordinates of the container above. For example, the host server **100** obtains the weather data of the day through a weather database S. The host server **100** estimates an expected power supply value of a photovoltaic device **500** based on absolute time, position coordinates, weather data, seasonal time,

dynamic changes of cloud clusters, solar elevation angle and azimuth angle, and the host server **100** obtains the expected power supply value through a comparison table. The comparison table includes historical variables and corresponding expected power supply values; then, the power supply value of the photovoltaic device **500** is compared with the expected power supply value, and a difference value is obtained; and a determination is made as to whether the difference value is within the expected range, if so, return to step **803**, otherwise, a smudge-detected alert will be issued outwardly.

(36) It is noted, the detailed options of the above step **804** may also be used simultaneously or sequentially.

(37) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(38) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the present disclosure. In view of the foregoing, it is intended that the present disclosure covers modifications and variations of this disclosure provided they fall within the scope of the following claims and their equivalents.

Claims

1. A smudge detection device, comprising: a power supply battery; a container comprising a main body disposed on at least one power generation module, a recess formed on a top of the main body, and a light transmissive cover covering the recess; a smudge detection module located within the recess for detecting a shielding degree of the light transmissive cover being covered by foreign matters; a photovoltaic device disposed on the main body for generating and transmitting electric energy into the power supply battery; a mainboard module received within the recess, comprising: a communication unit connected to a host server through a network architecture; and a control unit electrically connected to the smudge detection module, the power supply battery, the communication unit and the photovoltaic device, for respectively transmitting power supply value of the electric energy and the shielding degree of the light transmissive cover to the host server.
2. The smudge detection device of claim 1, wherein the light transmissive cover is a single light-transmissive plate, and the single light-transmissive plate is divided into a first area and a second area, wherein the photovoltaic device is directly integrated into the first area of the single light-transmissive plate, and the smudge detection module is faced towards the second area of the single light-transmissive plate from inside the recess for detecting the shielding degree of the second area being covered, wherein the shielding degree of the second area being covered is the same as the shielding degree of the light transmissive cover being covered.
3. The smudge detection device of claim 1, wherein the smudge detection module comprises: at least one outputting unit disposed within the recess, and electrically connected to the control unit for sending energy waves towards the light transmissive cover; and a receiving unit disposed within the recess, and electrically connected to the control unit for receiving reflected signals returned from the light transmissive cover for the control unit to transmit the reflected signals to the host server.
4. The smudge detection device of claim 3, wherein the at least one outputting unit comprises a plurality of outputting units, and the outputting units are spaced arranged to surround the receiving unit.
5. The smudge detection device of claim 3, wherein the at least one output unit is a light emitter, and the receiving unit is a light sensor; or the at least one output unit is an ultrasonic transmitter, and the receiving unit is an ultrasonic receiver.
6. The smudge detection device of claim 1, wherein the smudge detection module comprises: a

camera unit disposed within the recess, and electrically connected to the control unit for capturing images of the light transmissive cover for the control unit to transmit the images to the host server.

7. The smudge detection device of claim 1, wherein the mainboard module further comprises: a global positioning system (GPS) unit electrically connected to the control unit for obtaining location coordinates where the container is situated at, so that the control unit transmits the location coordinates of the container to the host server.

8. A smudge detection system, suitable for determining whether a surface of an observed object is smudgy, the smudge detection system comprising: a host server; and a smudge detection device comprising: a power supply battery; a container comprising a main body disposed on the observed object, a recess formed on a top of the main body, and a light transmissive cover covering the recess; a smudge detection module located within the recess for detecting a shielding degree of the light transmissive cover being covered by foreign matters; a photovoltaic device disposed on the main body for generating and transmitting electric energy into the power supply battery; and a mainboard module received within the recess, electrically connected to the smudge detection module, the power supply battery and the photovoltaic device, and connected to a host server through a network architecture, for respectively transmitting power supply value of the electric energy and the shielding degree of the light transmissive cover to the host server, wherein the host server determines whether the shielding degree of the light transmissive cover exceeds a preset threshold, and whether the power supply value of the photovoltaic device is within a predetermined range; and when the host server determines that the shielding degree exceeds the preset threshold, and the power supply value of the photovoltaic device is not within the predetermined range, the host server issues a smudge-detected alert.

9. The smudge detection system of claim 8, wherein the light transmissive cover is a single light-transmissive plate, and the single light-transmissive plate is divided into a first area and a second area, wherein the photovoltaic device is directly integrated into the first area of the single light-transmissive plate, and the smudge detection module is faced towards the second area of the single light-transmissive plate from inside the recess for detecting the shielding degree of the second area being covered, wherein the shielding degree of the second area being covered is the same as the shielding degree of the light transmissive cover being covered.

10. The smudge detection system of claim 8, wherein the smudge detection module comprises: at least one outputting unit disposed within the recess, and electrically connected to the mainboard module for sending energy waves towards the light transmissive cover; and a receiving unit disposed within the recess, and electrically connected to the mainboard module for receiving reflected signals returned from the light transmissive cover, wherein the mainboard module transmits the reflected signals to the host server, so that the host server is able to compare a difference between the reflected signal and the preset threshold.

11. The smudge detection system of claim 10, wherein the at least one outputting unit comprises a plurality of outputting units, and the outputting units are spaced arranged to surround the receiving unit.

12. The smudge detection system of claim 10, wherein the at least one output unit is a light emitter, and the receiving unit is a light sensor; or the at least one output unit is an ultrasonic transmitter, and the receiving unit is an ultrasonic receiver.

13. The smudge detection system of claim 8, wherein the smudge detection module comprises: a camera unit disposed within the recess, and electrically connected to the mainboard module for capturing images of the light transmissive cover for the mainboard module to transmit the images to the host server.

14. The smudge detection system of claim 8, wherein when the host server determines whether the power supply value of the photovoltaic device is within the predetermined range, the host server further proceeds: to collect an ideal power supply value of the photovoltaic device; to compare the power supply value of the photovoltaic device with the ideal power supply value to obtain a

difference value; to determine whether the difference value is in the predetermined range; and to issue the smudge-detected alert when the difference value is not within the predetermined range.

15. The smudge detection system of claim 8, wherein the mainboard module further comprises: a global positioning system (GPS) unit electrically connected to the mainboard module for obtaining location coordinates where the container is situated at, so that the mainboard module transmits the location coordinates of the container to the host server; and when the host server determines whether the power supply value of the photovoltaic device is within the predetermined range, the host server further proceeds: to obtain weather data and real-time sunshine data based on the location coordinates of the container, to estimate an expected power supply value based on the weather data and the real-time sunshine data; to compare the power supply value and the expected power supply value of the photovoltaic device to obtain a difference value; and to issue the smudge-detected alert when the difference value is not within the predetermined range.
